Fake News Detection in Social Media using Transformer Models

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1 Introduction

The fast growth of the social media has fundamentally changed the way in which information is produced and consumed. Without an editorial filter, social media lets anyone push content instantly, unlike the traditional journalism where there is an editorial filter, which forms fertile grounds of misinformation. Defined as fake news in general as a form of fabricated or intentionally misleading content that is still taken as a legitimate reporting, the problem has become a pressing issue in society. Its effects are real: faked news has distorted the minds of the population in the 2016 U.S. presidential election, and in the COVID-19 pandemic, the World Health Organization reported a parallel so-called infodemic of health misinformation that spreads at an even faster rate than the virus.

Initial methods of automated detection relied on manually-created linguistic features that were input to classical classifiers like Support Vector Machines and Naive Bayes models. These approaches were fragile and needed a lot of domain knowledge. Deep learning architectures - CNNs and LSTMs - enhanced the learning of representations significantly, but did it unidirectionally, which did not allow understanding the context. With the introduction of BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al., there was a turning point: pre-trained bidirectional self-attention provides deep word representations that are highly contextual and transfer potent information to the downstream classification task.

Leveraging this, Kaliyar, Goswami, and Narang suggested FakeBERT which is a hybrid architecture that adds parallel single-layer CNN blocks with varying kernel sizes to BERT embeddings to jointly capture both global semantic context and local n-gram patterns. On public benchmark datasets, the model has a classification accuracy of 98.90 percent, which is the best of all existing methods at the time. In this paper, FakeBERT is reviewed by outlining its methodology, critical gaps, and suggesting a problem statement to inform future research.

2 Methodology

FakeBERT is a hybrid deep learning system that fuses BERT's contextual representations with multi-scale convolutional feature extraction. The overall processing pipeline is illustrated in Figure 1.

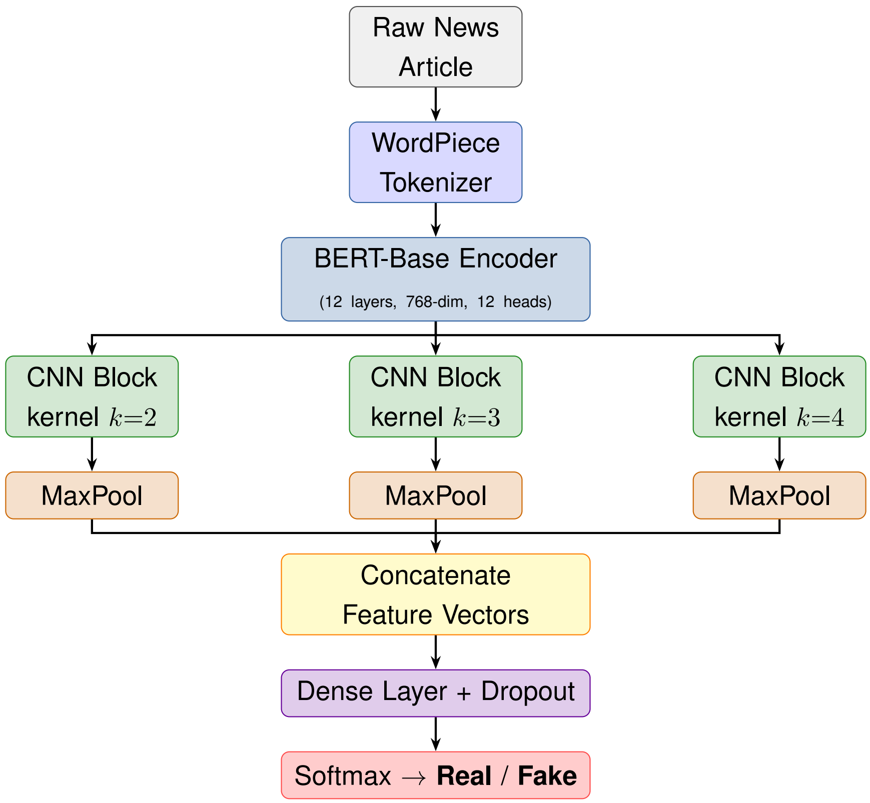

Figure 1: End-to-end pipeline of the FakeBERT model. Raw text is tokenized and encoded by BERT; three parallel CNN blocks with different kernel sizes extract multi-scale features; pooled outputs are concatenated and passed through a dense classifier.

2.1 Datasets

The authors evaluated FakeBERT on three public datasets, summarised in Table 1. The LIAR dataset contains ≈12,800 short political statements from PolitiFact with six veracity labels collapsed to binary. The ISOT dataset offers 44,898 full-length articles (real from Reuters, fake from flagged sites), providing rich textual content for deep feature extraction. FakeNewsNet spans political and entertainment news from GossipCop and PolitiFact; FakeBERT used only article text.

Table 1: Summary of datasets used in FakeBERT evaluation.

Dataset	Total Samples	Real	Fake	Domain
LIAR	12,836	6,321	6,515	Political statements
ISOT	44,898	21,417	23,481	News articles
FakeNewsNet	∼23,000	∼11,500	∼11,500	Political & Entertainment

2.2 Model Architecture and Training

Input text is tokenized with BERT's WordPiece tokenizer. A `[CLS]` token prepended to each sequence aggregates the sentence-level representation; a `[SEP]` token marks the sequence boundary. Sequences are truncated to 512 tokens. The pre-trained BERT-Base encoder (12 transformer layers, 768-dimensional hidden states, 12 self-attention heads) converts each token into a 768-dimensional context-sensitive vector. The `[CLS]` embedding, representing the whole sequence, is forwarded to the CNN stage.

Three parallel single-layer CNN blocks then process the BERT output simultaneously, using kernel sizes 2, 3, and 4. Each kernel slides over consecutive token embeddings, capturing k-gram level patterns — short phrase cues through broader contextual windows at the same time. ReLU activation introduces non-linearity; max-pooling retains the most salient features per filter. All pooled outputs are concatenated into one feature vector, passed through a fully connected layer with dropout, and finally through a softmax classifier. Training used binary cross-entropy loss with the Adam optimiser and an 80–20 train-test split.

2.3 Results

Table 2 compares FakeBERT against baselines. The model reaches 98.90% accuracy on ISOT — the richest textual dataset — and ≈89% on LIAR, outperforming all prior approaches in both cases. The ablation confirms that the parallel multi-kernel design is the key differentiator over a standalone BERT classifier.

Table 2: Accuracy (%) comparison of FakeBERT vs. baseline models.

Model	LIAR	ISOT	Precision	F1-Score
Naive Bayes	58.3	72.1	70.4	71.2
SVM (TF-IDF)	63.5	82.7	81.9	82.3
LSTM	71.2	91.4	90.8	91.1
CNN (single kernel)	74.8	93.2	92.7	93.0
BERT (standalone)	83.1	96.5	96.2	96.3
FakeBERT	**89.0**	**98.9**	**98.7**	**98.8**

3 Research Gap and Problem Statement

Despite strong benchmark results, FakeBERT leaves three important dimensions of the fake news problem unaddressed, visualised in Figure 2.

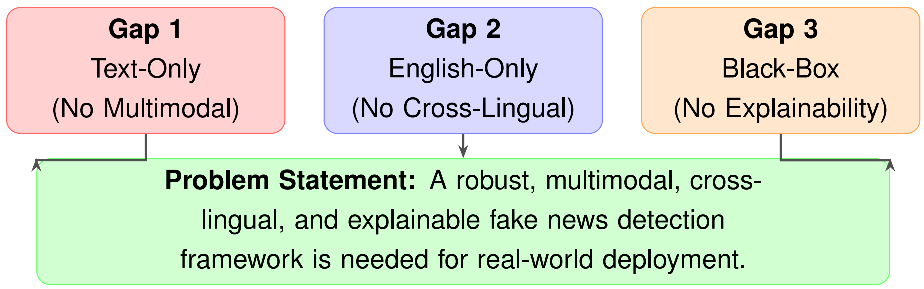

Figure 2: Three critical research gaps in FakeBERT and their convergence into the proposed problem statement.

Gap 1: Text-Only Design (No Multimodal Analysis). FakeBERT works with text only. Practically, much of social-media misinformation is distributed by use of images, videos and audio with deceptive text. Manipulated images are believed as a gut feeling and go viral. This type of fake news is structurally blind in a purely text-based model. The article recognizes the weakness but does not offer a solution.

Gap 2: English-Only Evaluation (No Cross-Lingual Generalization). Each of the three data sets is an English based corpus that focuses on American news. The model is not tested on other languages or other areas of culture. Counterfeit news is a worldwide issue - in most language low-resource communities, the fact-checking infrastructure is weak and the harm resulting in fake news is magnified. The patterns of rhetorical deception vary by culture as well, and thus a model trained with PolitiFact data might disastrously mislabel, e.g., Hindi or Arabic misinformation. The lack of crosslingual evaluation is a limiting factor in the real world use at scale.

Gap 3: Black-Box Classification (No Explainability). FakeBERT provides a one-bit label and does not give any rationale. The content moderation process, the journalistic verification process, and the policy enforcement process all necessitate that a human operator be aware of the purpose of the flagging of a piece of content. A system that does not give any attention map, saliency score, or natural-language explanation can hardly be trusted, is legally contestable, and has a tendency to silent failure modes. This is an especially acute gap in the light of increasing regulatory interest in algorithmic transparency.

Problem Statement

Current BERT-based fake news detection models such as FakeBERT have high accuracy on text-only English test sets, but are not applicable in practice because they cannot accept multimodal *inputs, do not generalize across languages and cultural settings, and lack interpretable decision outputs. The necessary support to facilitate real-world deployment is a solid, multimodal, cross-lingual, and explainable detection system, verified in a variety of environments, and able to offer clear explanations.*

4 Conclusion

FakeBERT is a real and highly motivated improvement to automated fake news detectors. Bidirectional contextual embeddings of BERT applied together with parallel multi-kernel CNN blocks result in a system more expressive than any of its components, with an accuracy of 98.90

There are however three research gaps that limit wider applicability: the text-only design omits a big and expanding category of visual misinformation; the English-only test has not been validated in global deployment; and the model lacks any mechanism to explain itself, which makes it ill-suited to responsible deployment on real-world data. Future research ought to seek collaborative textual-visual architectures, cross domain and multilingual assessments, and embedded explanations modules like attention visualization or rationale generation. No matter how precise automated detection tools are, they should be accompanied by media literacy campaigns and platform regulation to have a long-term effect on society. FakeBERT is a robust starting point, and the gaps here are the most obvious places that the community can develop it.

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